

Method of Images (as a consequence of Uniqueness Theorem)

Problem statement

A point charge q is held at a height d above an infinite conducting grounded plane (the plane $z = 0$). Find the electrostatic potential $\phi(x, y, z)$ in the region $z > 0$ (above the plane).

Challenge (physical): The point charge induces negative charges on the nearby surface of the conductor. We do *not* know beforehand how much charge is induced and how it is distributed on the plane.

Mathematical formulation (what we really solve)

In the region $z > 0$, the potential must satisfy Poisson's equation with a point source at $(0, 0, d)$:

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}, \quad \rho(\mathbf{r}) = q \delta(x) \delta(y) \delta(z - d), \quad (z > 0).$$

Boundary conditions (BCs) for a grounded infinite plane:

1. $\phi(x, y, 0) = 0$ for all (x, y) on the plane $z = 0$.
2. $\phi \rightarrow 0$ far away from the charge (e.g. $x^2 + y^2 + z^2 \rightarrow \infty$).

Why uniqueness theorem matters here

If we can construct *any* potential ϕ in $z > 0$ that (i) satisfies Poisson's equation for the given point charge and (ii) satisfies the BCs ($\phi = 0$ on $z = 0$, and $\phi \rightarrow 0$ at infinity), then by the **uniqueness theorem** that ϕ is the **unique** correct solution in $z > 0$.

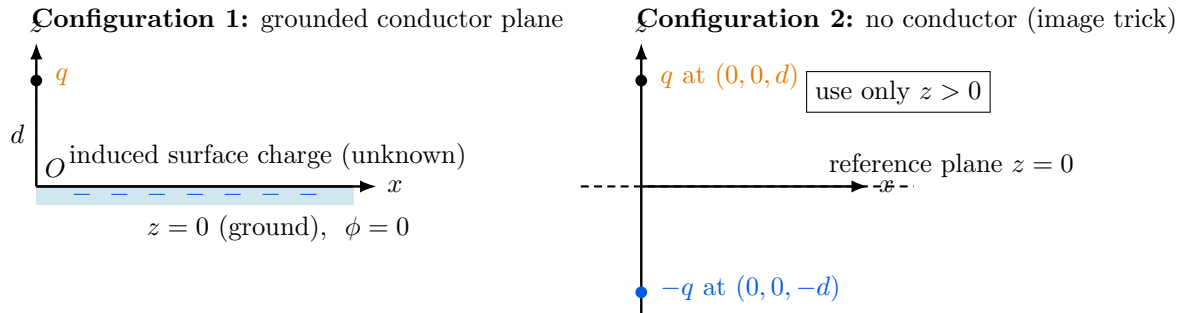


Figure 1: Idea of method of images: replace the grounded plane by an image charge $-q$ mirrored across $z = 0$. The resulting potential automatically satisfies $\phi = 0$ on $z = 0$.

The trick (method of images)

Instead of solving for the unknown induced charges on the grounded plane, we consider a **fictitious** configuration in free space: keep the real charge q at $(0, 0, d)$, and place an **image charge** $-q$ at $(0, 0, -d)$. Define distances

$$R_+ \equiv \sqrt{x^2 + y^2 + (z - d)^2}, \quad R_- \equiv \sqrt{x^2 + y^2 + (z + d)^2}.$$

Then by superposition of the electrostatic potential,

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_+} - \frac{q}{R_-} \right), \quad z > 0.$$

Check the boundary conditions (very important)

1. **Grounded plane $z = 0$:** At $z = 0$,

$$R_+ = \sqrt{x^2 + y^2 + d^2} = R_-,$$

so the two terms cancel:

$$\phi(x, y, 0) = 0.$$

2. **Far away:** As $r = \sqrt{x^2 + y^2 + z^2} \rightarrow \infty$, both $1/R_+$ and $1/R_-$ go to 0, hence

$$\phi \rightarrow 0.$$

Why this is the correct answer (connection to uniqueness theorem)

The above ϕ satisfies:

- Poisson's equation in the region $z > 0$ with the correct source (the real charge at $z = d$),
- $\phi = 0$ on the boundary plane $z = 0$,
- $\phi \rightarrow 0$ at infinity.

Therefore, by the **(Dirichlet) uniqueness theorem**, this ϕ is the **unique** correct potential in the region $z > 0$.

Takeaway (method of images)

To solve electrostatic problems with conductors:

- Replace the conductor boundary by suitable *image charges* so that the boundary condition (e.g. $\phi = 0$ on the conductor) is automatically satisfied.
- Then ϕ in the physical region is **guaranteed correct** by the **uniqueness theorem**.

For a charge q at $(0, 0, d)$ above a grounded infinite plane:

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right), \quad z > 0.$$

Induced surface charge on the grounded plane (from the image solution)

Potential used (method of images)

For a charge q at $(0, 0, d)$ above a grounded plane $z = 0$,

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right), \quad z > 0.$$

Let $\rho = \sqrt{x^2 + y^2}$. Then

$$\phi(\rho, z) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - d)^2}} - \frac{1}{\sqrt{\rho^2 + (z + d)^2}} \right).$$

Induced surface charge density $\sigma(\rho)$ on the conductor

Inside the conductor $z < 0$, the electric field is zero. Just above the surface $z = 0^+$, the boundary condition is

$$E_n(0^+) = \frac{\sigma}{\epsilon_0},$$

where E_n is the component normal to the conducting surface (here the normal is $\hat{\mathbf{z}}$ pointing into $z > 0$). Since $\mathbf{E} = -\nabla\phi$,

$$E_z(0^+) = - \left. \frac{\partial\phi}{\partial z} \right|_{z=0^+}.$$

Therefore,

$$\sigma(\rho) = \epsilon_0 E_z(0^+) = - \epsilon_0 \left. \frac{\partial\phi}{\partial z} \right|_{z=0^+}.$$

Compute $\partial\phi/\partial z$. Using

$$\frac{\partial}{\partial z} \left(\frac{1}{\sqrt{\rho^2 + (z \mp d)^2}} \right) = -\frac{(z \mp d)}{[\rho^2 + (z \mp d)^2]^{3/2}},$$

we get

$$\frac{\partial\phi}{\partial z} = \frac{q}{4\pi\epsilon_0} \left[-\frac{(z-d)}{(\rho^2 + (z-d)^2)^{3/2}} + \frac{(z+d)}{(\rho^2 + (z+d)^2)^{3/2}} \right].$$

Evaluate at $z = 0$:

$$\left. \frac{\partial\phi}{\partial z} \right|_{z=0} = \frac{q}{4\pi\epsilon_0} \left[-\frac{(-d)}{(\rho^2 + d^2)^{3/2}} + \frac{(+d)}{(\rho^2 + d^2)^{3/2}} \right] = \frac{q}{4\pi\epsilon_0} \left(\frac{2d}{(\rho^2 + d^2)^{3/2}} \right).$$

Hence

$$\sigma(\rho) = -\epsilon_0 \left. \frac{\partial\phi}{\partial z} \right|_{z=0} = -\frac{q d}{2\pi (\rho^2 + d^2)^{3/2}}.$$

Sign: $\sigma(\rho) < 0$, so the induced surface charge is negative (as expected for a positive charge q).

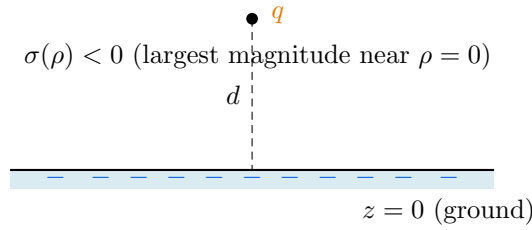


Figure 2: Induced surface charge is negative and concentrated near the point on the plane closest to the charge.

Total induced charge on the plane

The total induced charge is

$$Q_{\text{ind}} = \iint \sigma dS.$$

Use polar coordinates on the plane:

$$\rho = \sqrt{x^2 + y^2}, \quad dS = \rho d\rho d\theta.$$

Thus,

$$Q_{\text{ind}} = \int_0^{2\pi} \int_0^\infty \left(-\frac{q d}{2\pi(\rho^2 + d^2)^{3/2}} \right) \rho d\rho d\theta.$$

The θ -integral gives a factor 2π :

$$Q_{\text{ind}} = -q d \int_0^\infty \frac{\rho}{(\rho^2 + d^2)^{3/2}} d\rho.$$



Let $u = \rho^2 + d^2 \Rightarrow du = 2\rho d\rho$. Then

$$Q_{\text{ind}} = -q d \int_{u=d^2}^{\infty} \frac{1}{2} u^{-3/2} du = -q d \left[-u^{-1/2} \right]_{d^2}^{\infty} = -q d \left(0 - \frac{1}{d} \right) = \boxed{-q}.$$

Takeaway

For a point charge q at height d above a grounded infinite conducting plane:

$$\sigma(\rho) = -\frac{q d}{2\pi(\rho^2 + d^2)^{3/2}} < 0, \quad Q_{\text{ind}} = \iint \sigma dS = -q.$$

So the plane acquires a total induced charge $-q$, consistent with the image charge $-q$.

Force and Energy (method of images)

Force on the charge

A charge q above a grounded conducting plane is attracted toward the plane because the induced surface charge is negative. Using the method of images, the field in the region $z > 0$ is identical to the field produced by:

$$q \text{ at } (0, 0, d) \quad \text{and} \quad -q \text{ at } (0, 0, -d),$$

but we only use the field in the upper half-space $z > 0$.

Therefore the force on the real charge q equals the Coulomb force due to the image charge $-q$. The separation between the two charges is $2d$, so the magnitude is

$$F = \frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{4d^2}.$$

Direction is toward the plane (negative $\hat{z}$ direction), hence

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{z} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d^2} \hat{z}.$$

Energy: why we must be careful

Now consider the electrostatic energy.

Configuration 2 (free space, two point charges). If we truly had *two* point charges q at $z = d$ and $-q$ at $z = -d$ in free space, then the interaction energy would be

$$W_2 = \frac{1}{4\pi\epsilon_0} \frac{q(-q)}{2d} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}.$$



Energy: deriving the interaction term using $W = q \phi$

A clean way to compute the *interaction* energy is to *assemble* the charges from infinity.

Assembly argument (two point charges in free space). Place

$$q_1 = +q \text{ at } z = +d, \quad q_2 = -q \text{ at } z = -d, \quad R = |(+d) - (-d)| = 2d.$$

1. **Bring $q_1 = +q$ from infinity to $z = +d$.** At this stage there is no other charge present, so the potential at the arrival point due to *other charges* is zero:

$$\phi_{\text{others}}(z = +d) = 0 \implies W_1 = q_1 \phi_{\text{others}}(z = +d) = 0.$$

2. **Bring $q_2 = -q$ from infinity to $z = -d$.** Now the charge q_1 is already in place, so the potential at $z = -d$ due to q_1 is

$$\phi_1(z = -d) = \frac{1}{4\pi\epsilon_0} \frac{q_1}{R} = \frac{1}{4\pi\epsilon_0} \frac{q}{2d}.$$

Hence the work done by the external agent to bring q_2 in quasistatically is

$$W_2 = q_2 \phi_1(z = -d) = (-q) \left(\frac{1}{4\pi\epsilon_0} \frac{q}{2d} \right) = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}.$$

Therefore, the total *interaction* energy (assembly work, excluding self-energies) is

$$W_{\text{int}} = W_1 + W_2 = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d},$$

which is exactly the Coulomb interaction energy for two charges separated by $2d$.

Why we must be careful (self-energy and double counting)

- For a point charge, its own potential at its own location is singular, so a naive use of $W = q \phi$ with ϕ including “self” gives an unphysical infinite term (the self-energy).
- A safe summary formula for the *interaction* energy is

$$W_{\text{int}} = \frac{1}{2} \sum_i q_i \phi_{\text{others}}(\mathbf{r}_i),$$

where $\phi_{\text{others}}(\mathbf{r}_i)$ is the potential at $\mathbf{r}_i$ due to all charges *except* q_i . The factor $1/2$ avoids double counting of pair interactions.

But our physical problem is Configuration 1 (grounded plane). In the real grounded-plane problem, only the region $z > 0$ contains nonzero field and stores energy. By symmetry, the field-energy of

the two-charge free-space configuration is split equally between $z > 0$ and $z < 0$. Hence the energy stored in the *physical* problem is half of W_2 :

$$W_1 = \frac{1}{2} W_2 = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}.$$

(In the class we highlights this key idea: energy in Configuration 1 is $\frac{1}{2}$ of the energy of Configuration 2.)

Important caution (as discussed in class)

Although the *field* in $z > 0$ is identical to that of the two-charge configuration, the **total energy** is *not* the same, because the physical region is only $z > 0$. So $W_1 \neq W_2$; rather $W_1 = \frac{1}{2} W_2$.

Alternate method to compute energy (work done from infinity)

Another clean way is to compute the work required to bring q from infinity to height d .

At a general height z , the force magnitude is

$$F(z) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{(2z)^2} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{4z^2},$$

directed downward (toward the plane). To bring the charge in *quasi-statically*, the external agent must apply an equal and opposite force, so the required work is

$$W_1 = \int_{\infty}^d \mathbf{F}_{\text{ext}} \cdot d\mathbf{l} = \int_{\infty}^d \left(+\frac{1}{4\pi\epsilon_0} \frac{q^2}{4z^2} \hat{\mathbf{z}} \right) \cdot (\hat{\mathbf{z}} dz) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{4} \int_{\infty}^d \frac{dz}{z^2}.$$

Evaluate:

$$\int_{\infty}^d \frac{dz}{z^2} = \left[-\frac{1}{z} \right]_{\infty}^d = -\frac{1}{d}.$$

Hence

$$W_1 = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}.$$

This matches the $\frac{1}{2} W_2$ result above.

Intuition: why energy is half (Configuration 1 vs 2)

- In **Configuration 1** (grounded conducting plane), while bringing q closer, we do work only on the real charge q . The induced charges move on the conductor surface, but the conductor is an equipotential at $\phi = 0$, so no additional “separate work” is counted for them.
- In **Configuration 2** (two point charges in free space), to create the same final arrangement, work must be done on *both* charges (symmetrically). That is why the total energy there is twice the energy stored in the physical half-space problem.

